Wallet Transfer Flow

Endpoint

POST /api/v1/transfers

Requirements:

- valid JWT;
- active durable session matching JWT sid;
- Idempotency-Key;
- source wallet owned by JWT sub customer;
- distinct destination wallet;
- positive DECIMAL(19,4)-compatible amount;
- matching source/destination currency;
- final fraud Allow;
- sufficient source balance.

```
{
  "sourceWalletId": "11111111-1111-1111-1111-111111111111",
  "destinationWalletId": "22222222-2222-2222-2222-222222222222",
  "amount": 125.50
}
```

Currency is not trusted from the client; it is derived from wallet state.

Idempotency order

Completed replay is checked **before** fraud:

```
request
-> completed replay lookup
   -> same key + same immutable payload: original result
   -> same key + different payload: conflict
-> durable session/risk
-> internal fraud
-> external fraud
-> atomic posting
```

This precheck is a semantic/performance guard. Final correctness still belongs to Serializable idempotency locking inside the atomic posting store.

Session validation

A valid JWT alone is not sufficient for money movement. sid must:

- belong to the same customer;
- not be revoked;
- not be expired;
- belong to an Active customer session.

Server-derived fraud signals

The client does not submit risk flags. SqlWalletTransferRiskSignalStore derives from server state:

- customer country;
- wallet currency;

- device reference/history;
- new-device flag;
- five-minute transfer velocity;
- 24-hour same-currency amount;
- known beneficiary.

Raw DeviceId is not sent to the provider; a stable SHA-256 reference is used.

Fraud flow

Internal Deny cannot be overridden by external Allow. FakeFraud is required after internal Allow/Review. External timeout/network/malformed response fails closed.

```
Internal + External -> FraudDecisionPolicy -> Allow / Review / Deny
```

- Allow -> posting;
- Review -> no money movement;
- Deny -> no money movement;
- provider unavailable -> no money movement.

A durable manual-review queue is not yet implemented.

Atomic posting

After final Allow, one MSSQL transaction commits:

- IdempotencyRecord;
- source debit;
- destination credit;
- FinancialTransaction;
- Wallet Liability LedgerAccounts;
- balanced LedgerJournal + LedgerEntries;
- persisted SQL Debit/Credit equality validation.

External HTTP never runs inside this transaction.

Double-entry

```
Debit   source wallet liability       amount
Credit  destination wallet liability  amount
```

Journal balance is validated at both Domain and SQL levels.

Replay response

Replay contains only immutable fields:

- FinancialTransaction ID;
- source/destination wallet IDs;
- amount/currency;
- original completion time;
- replay flag.

Current wallet balances are excluded because they may have changed since the original transaction.